Science - K-12 Level Assessment 2 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. A car is rolling on a level street. What can make the car stop moving?
   (a) a force that pushes straight up
   (b) a force that pushes straight down
   (c) a force that pushes the same direction that the car is moving
   (d) a force that pushes the opposite of the direction that the car is moving

2. The human body produces motion by changing chemical energy into mechanical energy. Which of these best describes what happens to the energy?
   (a) The total amount of energy increases.
   (b) The total amount of energy is constant.
   (c) The energy is destroyed through motion.
   (d) The amount of chemical energy increases.

3. A student walking through the forest notices moss growth increases as the density of the forest increases. The student has made
   (a) a hypothesis.
   (b) a conclusion.
   (c) an assumption.
   (d) an observation.

4. The gallbladder stores bile, a fluid that aids in digestion of fat. Which of the following types of food should be avoided by a person whose gallbladder has been removed?
   (a) fruits
   (b) grains
   (c) cheese
   (d) vegetables

5. A strong magnet will separate a mixture of
   (a) clear glass and green glass.
   (b) paper cups and plastic cups.
   (c) iron nails and aluminum nails.
   (d) sand and salt.

True / False (10 marks)

Answer True or False

6. True or False: Renewable resources will be available for many years, providing a sustainable energy supply.
7. True or False: Earth's tilt on its axis causes one side to receive less energy from the Sun during December.
8. True or False: Stems transport water to other parts of the plant through a system of tubes called xylem.
9. True or False: The Alps, Appalachians, and Himalayas are formed primarily from folded rock due to tectonic forces.
10. True or False: The Big Bang theory states that the universe started as a single mass that expanded over time.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process of photosynthesis in plants, specifically focusing on how sunlight is converted into energy and the significance of this transformation for the plant and the ecosystem.
12. Discuss the importance of labeling in scientific experiments, particularly in plant growth studies, and explain how it contributes to the reliability of data for other researchers.

End of Paper